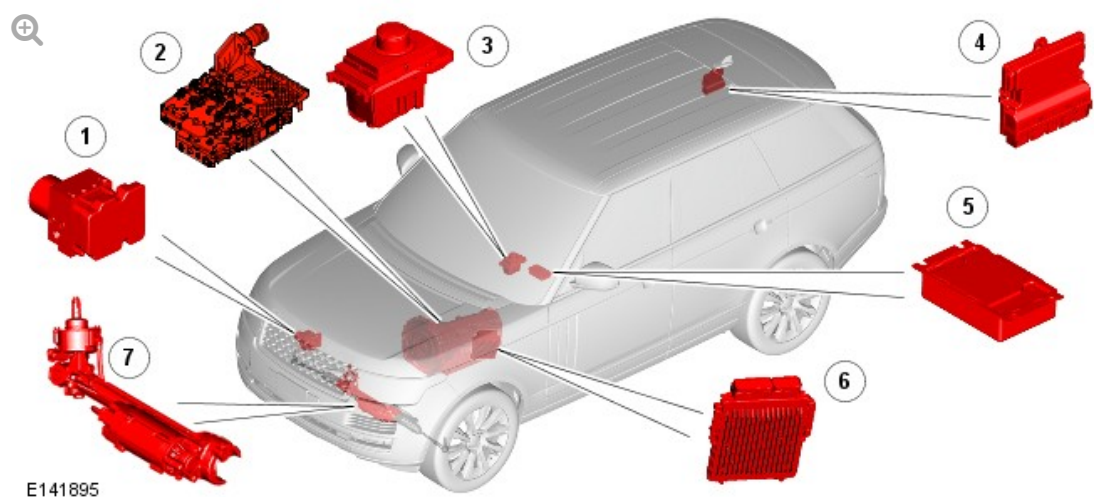


2016.0 RANGE ROVER (LG), 204-06

RIDE AND HANDLING OPTIMIZATION

DESCRIPTION AND OPERATION

TERRAIN RESPONSE



E141895

ITEM	DESCRIPTION
1	ABS module
2	Transmission control module
3	Terrain Response Switchpack
4	Integrated Suspension Control Module (ISCM)
5	Transfer Box Control Module
6	Engine control Module (ECM)
7	Power Steering Control Module (PSCM)

OVERVIEW

The Terrain Response® system allows the driver to select a program which aims to provide the optimum settings for traction and performance for the prevailing terrain conditions. The system cannot be switched off. The default 'general program' is an active program that covers all general driving conditions. Seven specific terrain programs are selectable to cover all terrain surfaces.

The system is controlled by a Terrain Response® switchpack. The Terrain Response® switchpack allows the selection of one of the following seven programs:

- Dynamic program
- General program (special programs off)
- Grass, Gravel, Snow program
- Mud-Ruts program
- Sand program
- Rock Crawl program
- Automatic program

The instrument cluster displays the selected program in the message center excluding 'Automatic program'.

The Terrain Response® system uses a combination of a number of vehicle sub-systems to achieve the required vehicle characteristics for the terrain selected. The following sub-systems are used in the Terrain Response® system:

- Transfer Case Control Module
- Transmission Control Module
- Engine Control Module

- Power Steering Control Module
- ABS Module
- Integrated Suspension Control Module

DESCRIPTION

Each sub-system operates in different ways in relation to the selected Terrain Response® program to achieve the optimum traction, stability and ease of control for the terrain encountered.

ENGINE MANAGEMENT SYSTEM

The ECM changes the accelerator pedal maps to modify the amount of torque per percentage of pedal travel.

Each Terrain Response® program uses a combination of operating parameters for each sub-system. Changing between terrain programs initiates a different set of operating characteristics which will be noticeable to the driver, for example; if the accelerator pedal is held in a constant position and the terrain program is changed from Grass/Gravel/Snow to Sand, the driver will notice the torque and engine speed increase. If the terrain program is changed from Sand to Grass/Gravel/Snow, the driver will notice a reduction in torque and engine speed.

TRANSMISSION CONTROL

The TCM changes the shift maps for the Terrain Response® program selected. This changes the shift points providing early or late upshifts and downshifts when compared to general program.

For example, in the sand program, the transmission will perform later upshifts and earlier downshifts to maintain a higher engine speed.

Sport mode is only available when general program is selected. Sport mode is disabled in all Terrain Response® special programs. When S (sport) is selected on the transmission selector in a Terrain Response® special program, the transmission will select Command Shift™. Sport mode will not

be available.

TRANSFER CASE AND REAR DIFFERENTIAL CONTROL

The Transfer case electronically controlled differential and the rear electronically controlled differential (if fitted) are treated as one system. The electronic rear differential is an optional fitment on vehicles fitted with the Terrain Response® system. The differential control has two operating strategies; pre-emptive and reactive.

The pre-emptive strategy anticipates and predicts the locking torque value required for each differential to minimize slip and maximize stability. Each Terrain Response® program has a different threshold and input criteria for the pre-emptive strategy. The pre-emptive strategy improves vehicle traction and composure by avoiding wheel spin. This is achieved by anticipating the amount of differential lock required for the program selected. For example, a high locking torque would be applied for rock crawl or slippery surfaces.

The reactive strategy varies the amount of locking torque in response to the actual slip level and the dynamic behavior of the vehicle. Each Terrain Response® program has a different threshold and input for the reactive strategy. The reactive strategy improves vehicle traction and composure by eliminating any wheel spin which has occurred after the pre-emptive strategy was applied. The locking response applied is applicable to the terrain program selected, for example, very sensitive on slippery surfaces to provide maximum traction and minimize surface damage.

The locking torque calculations use various signals from other subsystems, for example, engine torque, throttle position, selected gear, steering angle, vehicle speed, lateral acceleration, yaw behavior.

The Dynamic Stability Control function of the ABS system can override the Terrain Response® differential control and reduce any applied locking torque during DSC action.

For additional information, refer to: [Braking Control System](#) (206-11 Brake Controls, Description and Operation).

ABS SYSTEM CONTROL

The ABS module controls several functions and adjusts the operating parameters of these functions to optimize the selected Terrain Response® program.

TRACTION CONTROL

Traction control uses different slip/acceleration thresholds to improve traction and vehicle composure. For example, the system sensitivity is increased on slippery surfaces such as wet grass or snow to reduce wheel spin. If wheel spin was allowed in these circumstances, loss of traction may result from surface damage (wet grass) or the car being unable to move (snow) due to wheel spin.

DYNAMIC STABILITY CONTROL

If DSC is switched off (with the DSC switch) when using a Terrain Response® special program, if the special program is subsequently changed for a different program, DSC is automatically switched back on.

DSC uses different threshold values for the selected Terrain Response® program to minimise DSC intervention, removing the requirement for the driver to disable the DSC system in order to reduce engine intervention which is sometimes induced in extreme off-road conditions. In extreme sand conditions, there may be an additional benefit of disabling the DSC function using the DSC switch in addition to selecting the sand program.

In the mud/ruts program the DSC system is calibrated to tolerate a higher yaw threshold. This allows a greater differential between the actual and desired wheel turning behavior before DSC intervenes. This allows the DSC system to ignore the effect of ruts 'jarring' the car or adjusting the front wheel steering angle.

The ABS system can alter the balance between engine and brake intervention.

HILL DESCENT CONTROL (HDC)

HDC is automatically switched on or off and target speeds are adjusted in

response to the Terrain Response® program selected. The responsiveness of the HDC function is also increased where required.

Automatic selection of HDC aims to assist the driver by switching the system on or off when it is of most benefit. HDC is only automatically switched on when the mud/ruts program is selected. HDC is automatically switched on when the following programs are selected:

- Mud-Ruts program (high range selected)
- General program (low range selected)
- Grass, Gravel, Snow program (low range selected)
- Mud-Ruts program (low range selected)
- Rock Crawl program (low range selected)
- Automatic program (low range selected)
- Mud-Ruts program (low range selected)

ADAPTIVE DYNAMICS CONTROL SYSTEM

In dynamic program, the suspension delivers tighter body control, with flatter handling and sharper responses.

POWER STEERING SYSTEM CONTROL

The power steering module changes the steering maps for the Terrain Response® program selected. This will change the active steering (self-centering) and active damping functions of the steering system when compared to special programs off. For example, in sand program the steering will exhibit greater self-centering torque to overcome the increased drag when driving in deep soft sand. This is to reduce the overall driving efforts and provide improved road-wheel feedback characteristics when used in such conditions.

INCORRECT PROGRAM USAGE

Selection of an inappropriate program is discouraged in the following ways:

- The Terrain Response® control module 'locks out' certain functions in

some programs, for example:

- Cruise control is only available with the general program or grass/gravel/snow program.
- Sport mode is deactivated in all special Terrain Response® programs. The vehicle will automatically select Command Shift™ when the vehicle is in a Terrain Response® special program.

Selection of an inappropriate program for the terrain conditions will not endanger the driver or cause damage to the vehicle. Continued use of an inappropriate program may reduce the life of some components. The driver may notice reduced vehicle response, with the engine and transmission being less responsive than in general program. Also, in some programs, HDC will remain on, signified by illumination of the HDC indicator in the instrument cluster.

DRIVER INFORMATION

The message center in the instrument cluster contains the Terrain Response® program icons which display the currently selected program. No symbol is displayed when auto program off is selected.

Any required changes to the sub-systems are also passed to the driver in the form of indicator illumination in the instrument cluster or appropriate messages in the message center, HDC OFF for example.

In certain operating conditions, the Terrain Response® system also displays advice or warning messages to ensure the driver is using the vehicle to its full potential, for example, steering angle is displayed in the message center to avoid driving in deep ruts with steering lock applied.

DIAGNOSTICS

The Terrain Response® control module stores information on detected Terrain Response® faults and CAN (controller area network) errors which can be interrogated using approved diagnostic equipment. The Terrain Response® sub-systems and the instrument cluster also store information relating to CAN errors from the Terrain Response® control module.

The control module also stores the distance traveled and time elapsed for the individual programs which can also be retrieved using approved diagnostic equipment. This information aids diagnosis of the Terrain Response® system and also provides an indication of Terrain Response® system abuse by the driver which can lead to premature component failure. This information can also be used to check customer concerns, for example, high fuel consumption which may be due to continued use of a certain program.

TERRAIN RESPONSE® SYSTEM FAULT DIAGNOSTICS

Terrain Response® relies on the correct functionality of the sub-systems. If one of the sub-systems develops a fault, the Terrain Response® system will not function, even though the fault is not in the Terrain Response® system.

It is not possible for the Terrain Response® control module to cause any fault behavior (warning indicator illumination or message generation) in any of the sub-systems. Illumination of a sub-system warning indicator and/or a sub-system related message will never be associated with a Terrain Response® control module or Terrain Response® system fault.

The Terrain Response® switchpack should only be investigated if there are no apparent faults in any of the sub-systems. If a fault in a sub-system is subsequently corrected, the Terrain Response® system will function normally after an ignition cycle.

TERRAIN RESPONSE® SUB-SYSTEM FAULTS

When a fault occurs in a sub-system, the driver is alerted by the illumination of a warning indicator and/or an appropriate message for that sub-system in the instrument cluster message center. There will be no warning of a Terrain Response® system fault.

When a sub-system fault is present and the driver attempts to select a different Terrain Response® program, or at the next ignition mode 2 (on), the message 'TERRAIN RESPONSE SPECIAL PROGRAMS NOT AVAILABLE' will appear in the message center. This implies that the Terrain Response® system has a fault, but only because a sub-system fault is preventing its

operation. This message will be displayed for 5 seconds per power mode 6 (ignition on) occurrence, but is repeated if a further selection is made by the driver using the Terrain Response® switchpack or at the next power mode 6 occurrence. When a sub system fault is present , the Terrain Response® switchpack will show a general program as selected.

NOTE:

The message 'TERRAIN RESPONSE SPECIAL PROGRAMS NOT AVAILABLE' can also be generated by a fault in the Terrain Response® switchpack or control module. Refer to the following information for details of switchpack or control module faults.

It is not possible for the Terrain Response® control module to cause any fault behavior (warning indicator illumination or message generation) in any of the sub-systems. Illumination of a sub-system warning indicator and/or a sub-system related message will never be associated with a Terrain Response® control module or Terrain Response® system fault.

TERRAIN RESPONSE AUTO2® SWITCHPACK AND CONTROL MODULE FAULTS

If a fault occurs in the Terrain Response® switchpack, the amber LED (light emitting diode) in all of the switchpack icons are turned off and program selections are ignored. The instrument cluster message center displays the message 'TERRAIN RESPONSE SPECIAL PROGRAMS NOT AVAILABLE' when the fault occurs, if the fault is present and the driver attempts to select a special program (if the control module is able to do this) or at the next ignition cycle.

If a CAN fault exists and prevents Terrain Response® system operation, general program only is illuminated. When the driver tries to select another program they will get the system fault message.

If the instrument cluster does not receive a Terrain Response® system CAN message from the Terrain Response® control module, the message

'TERRAIN RESPONSE SPECIAL PROGRAMS NOT AVAILABLE' will be displayed when the fault occurs and will be repeated at every ignition mode 2 (on).

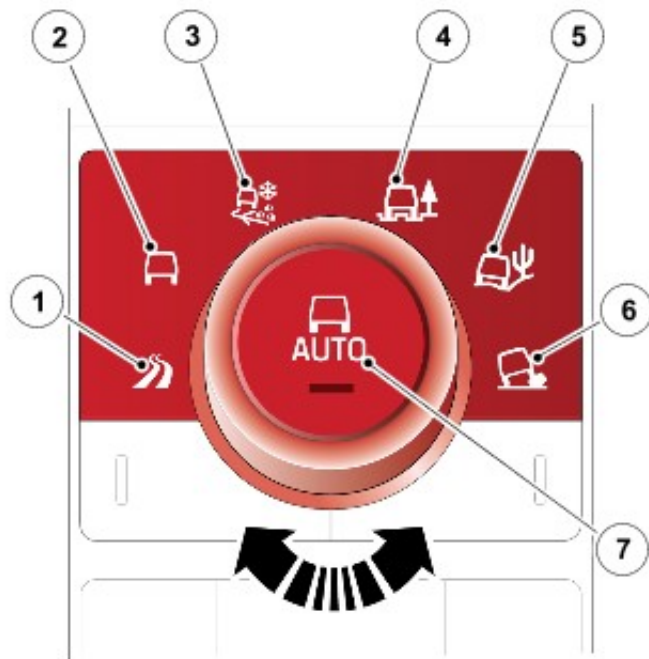
USER ERROR

The following incorrect usage of the system may be misinterpreted as a system fault:

- Engine not running - Program changes and driver advisory messages are only available with the engine running.
- Special program change attempted with DSC or ABS active (this includes ABS cycling which is operational when HDC is being used on slippery or loose surfaces).
- Special program change attempted with overheat condition present on the active on-demand coupling.

COMPONENT DESCRIPTION

TERRAIN RESPONSE® SWITCHPACK AND CONTROL MODULE



E154931

ITEM	DESCRIPTION
1	Dynamic program
2	General
3	Grass, Gravel or Snow
4	Mud-Ruts
5	Sand
6	Rock Crawl
7	Automatic

The switchpack is installed in the floor console immediately behind the gear selector. Each program is denoted by a symbol that represents the related terrain. Programs are selected using the rotary control switch to illuminate the required program indicator.

The switchpack is connected to the vehicle wiring with a harness connector

that contains connections for the illumination supply, ignition supply, ground, and high speed CAN bus (positive and negative).